

Lesson: Module 3 - Run an A/B/C Test & Pick a Winner

Video: M3V3 – Demo: Running Your First A/B/C Benchmark
Purpose: Run three model variants on a fixed evaluation set to capture TTFT, latency, throughput, cost, and quality — then visualize trade-offs to pick a winner.
Commands:

```
# Navigate to project

cd abc_demo

# Set OpenAI API key

export OPENAI_API_KEY="your-key-here"

# Run Variant A (baseline)

python run_variant.py --model gpt-4o-mini --variant A --output results_A

# Run Variant B

python run_variant.py --model gpt-4o-mini --variant B --output results_B

# Run Variant C

python run_variant.py --model gpt-4o-mini --variant C --output results_C
```

Key Files to Open:

- eval_set.jsonl – Fixed evaluation prompts (10 tasks)
- run_variant.py – Benchmarking harness with streaming + timing
- results_A/raw.jsonl – Detailed per-request logs for Variant A
- results_A/summary.csv – Clean metrics for analysis
- results_B/ and results_C/ – Same structure for other variants
- analyze_results.ipynb – Pandas + matplotlib comparison notebook

Key Metrics Captured:

- TTFT (time-to-first-token): 0.57s – 0.62s (p50)
- p95 TTFT: 0.97s – 1.26s
- Total latency: 3.3s – 3.66s (p50)
- Tokens per second: 35 – 38
- Cost per task: \$0.000059 – \$0.000062
- Quality score: 1.0 (all variants)